

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460731.846 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.021
Transit depth (Rp/Rs)^2	2.81 +/- 0.71 [%]
Orbital Inclination (inc)	81.03 +/- 1.76
Airmass coefficient 1 (a1)	1.162 +/- 0.014
Airmass coefficient 2 (a2)	-0.102 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	None
Optimal Aperture	4.31
Optimal Annulus	6.46
Transit Duration (day)	0.0468 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.168 ± 0.021 vs 0.1594 (deviation 0.41σ)
Duration fit vs published	0.0468 d vs 0.0512 d (deviation 0.56σ)
Mid-time O-C	-4.1 min
Depth SNR	8.0
Residual scatter	1.25 %

Light curve

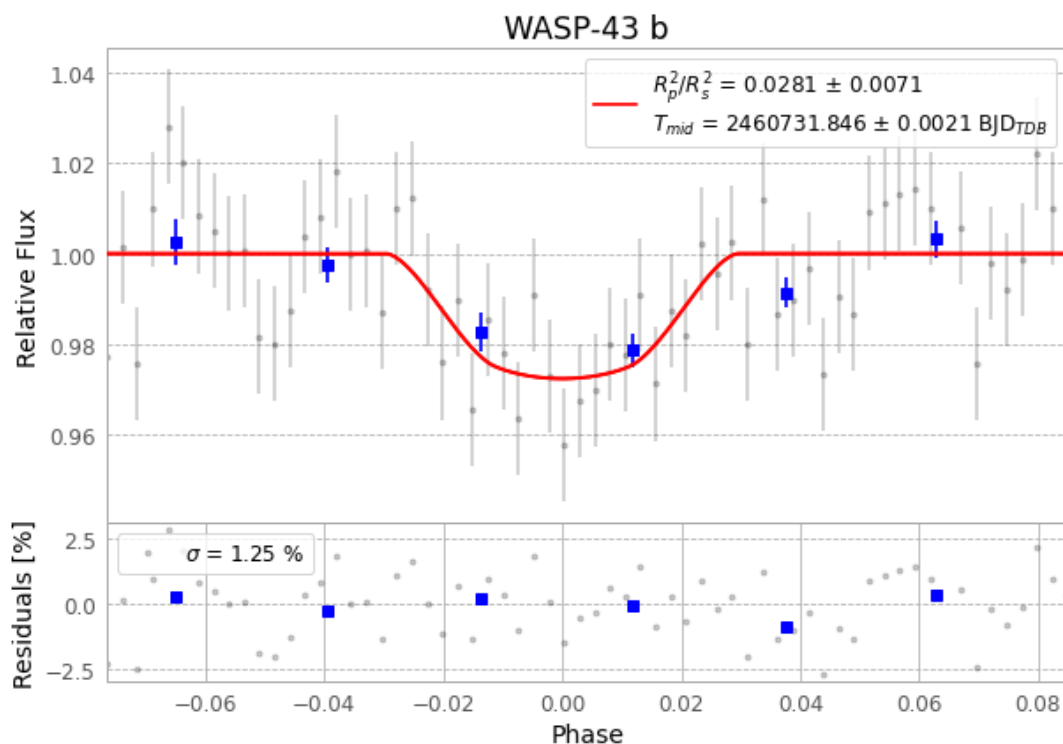


Figure 1 — FinalLightCurve_WASP-43 b_2025-02-24.png (SHA-256 7f1e8d1e9272c976...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 249], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 872ba9448c26acd3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

64 FITS frames (42 MB), WASP-43250225063915.FITS → WASP-43250225094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250225.log (529658f06d25...), FinalLightCurve_WASP-43 b_2025-02-24.png (7f1e8d1e9272...), FinalLightCurve_WASP-43 b_2025-02-24.csv (c37b3ca375e4...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 02:16Z.